# Topic: Discrete Fourier Transform (DFT–FFT) and Audio Denoising

## Required Content

1. Build the theoretical foundation of the Fourier transform:
   - From Fourier series (continuous case) to the Fourier transform;
   - Discretization of signals and derivation of the Discrete Fourier Transform (DFT);
   - The idea of reducing computational complexity using the Fast Fourier Transform (FFT).
2. Analyze and explain the quantities in the DFT formula:

$$X_k = \sum_{n=0}^{N-1} x[n] e^{-2\pi i k n/N}.$$

   - Meaning of $x[n]$, $X_k$, $|X_k|$, and the index $k$;
   - Time domain and frequency domain concepts;
   - Differences between the two domains in the context of audio denoising.
3. Apply the method to the audio denoising problem:
   - Transform the signal to the frequency domain;
   - Identify and remove (or attenuate) noise components;
   - Apply the inverse Fourier transform to reconstruct the signal.

## Description

- **Object**: Real audio files (recordings with noise; preferably monophonic or low-polyphony signals).
- **Purpose**: Understand the mathematical nature of DFT–FFT and their practical significance in audio signal processing.
- The implementation process consists of the following steps:
  - Constructing the theoretical background;
  - Implementing DFT/FFT algorithms;

  - Spectrum analysis;
  - Noise filtering;
  - Signal reconstruction.

## Expected Results

- Develop a program that implements DFT (and/or FFT) directly from the formulas.
- Illustrate audio waveforms in the time domain before and after denoising.
- Illustrate frequency spectra before and after noise filtering.

## Notes

- Do not directly use built-in functions such as `fft()` or `ifft()` to compute the main results.
- Clearly present the algorithmic idea before implementation.